Practice Tests 1

Course Name: Database Principle Examination Time: 120minutes
Student Number:___________________Student Name:___________________

1. What is database integrity? How is it different from database security? (5+5)

2. What is the basic principle of database recovery? What is the implementation of database recovery? (4 + 6)

3. The relations R and S are shown in the figure below. Please provide the results of the following expressions.（5+10）

R

A	B	C	D
a_1	b_1	c_1	d_1
a_1	b_1	c_2	d_2
a_1	b_1	c_1	d_3
a_2	b_2	c_1	d_1
a_2	b_2	c_2	d_2
a_3	b_3	c_1	d_1

S

C	D
c_1	d_1
c_2	d_2

(1) Calculate the relational algebra operation: $\Pi_{A,C,D}(_{A!=a1}(R)) \div S$

(2) Express the division operation using domain relational calculus.

4. Please answer the following questions:

With the coming of the World Cup, how to manage football activities has become a hot topic. In order to solve this problem, we built a <u>Football Activity Management System</u>, which needs to maintain the information of the <u>Team</u>, <u>Player</u>, <u>Chief Coach</u>, <u>Chief Referee</u>, <u>Sponsor</u>, And each <u>Match</u>. The data schema is set as follows:

[1] Team (TeamId, TeamName, PlayersNum, EstablishDate, ChiefCoach) ;

[2] Player (PlayerId,PlayerName, BirthDate, Height, Weight) ;

[3] Chief Coach (CoachId,CoachName, BirthDate, CoachLevel) ;

[4] Chief Referee (RefereeId,RefereeName, BirthDate, RefereeLevel) ;

[5] Sponsor (SponsorId, SponsorName)

[6] Match (MatchId, HomeTeamId, VisitTeamId, ChiefRefereeID, MatchDate, Score).

Assume that the football system meet the following arrangements:

[1] Each team has only one chief coach and several players.

[2] Each chief coach can only be hired by one team, and each player only play for one team.

[3] There are two teams in each match, one for the home team and one for the visiting team. There is only one chief referee for each match. Each match has a unique id and the date and score of the match are recorded.

[4] Each sponsor can sponsor multiple teams, but each team can only have one sponsor. Each sponsor can individually sponsor multiple players, and each player can also represent multiple sponsors.

[5] Additionally, it allows the cases where one team plays more than one matches in the same day or plays with the same team in the same day.

(1) Please draw the ER diagram of the football activity management system according to the above Settings (5).

(2) Please write Primary Key,Candidate Key and Foreign Key of the Match table(10).

5. Please answer the following questions:

The relationship between Teachers and Courses is shown in the following table:

Course ID(key)	Teacher Name	Department ID
C1	David	D1
C2	Mally	D2
C3	Musk	D1
C4	Musk	D1

(1) Which paradigm does the above relationship belong to? Why? (5)

(2) Does there exist the problem of "Delete Exception" or "Modify Exception"? If it does, how do they happen? (5)

6. Please answer the following questions:

It is known that the data relationship of local enterprises is as follows:

[1] Information of Employees: EMP(Eno, Ename, Sex, Age, Title)

[2] Information of Companys：CPY(Cno, Cname, Area)

Please answer the question according to the above information:

(1) Please CREATE an information table WORKIN where employees work in the enterprise and meet the following conditions: (10)

[1] The WORKIN table contains three attribute (the Eno from Emp, the Cno from CPY and the Salary)

[2] Define the primary and foreign key.

[3] Set a constraint that the Salary should not be less than 3,000 yuan.

(2) Based on the WORKIN table created in question (1), fill in the following SQL statement implementation function:

a) Query the Cno and Cname of the company which has the most employees (2+3)

```
SELECT C.Cno,C.Cname
FROM Company AS C， WORKIN AS W
WHERE C.Cno = W.Cno
GROUP BY ____(A)____
HAVING ____(B)____  （SELECT COUNT(*)
                       FROM WORKIN AS W2
                       GROUP BY W2.Cno）
```

b) Query the Cno and Area of the Company whose employees have the oldest average Age (2+3)

SELECT C.Cno，C.Area
FROM EMP AS E，CPY AS C，WORKIN AS W
WHERE W.Cno = C.Cno
AND W.Eno = E.Eno
GROUP BY C.Cno, C.Area
HAVING ____(C)____ >= ____(D)____

(3) Please fill in the vacancy part of the following TRIGGER to realize the automatic maintenance function of employee salary. When the Title of the employee changes, the salary of the new title should be calculated according to the "Title-Salary" Function——SalaryList(Title) . And the salary should be updated automatically: (1+2+2)

CREATE TRIGGER NewSalary
AFTER UPDATE ____(E)____ON EMP
REFERENCING NEW AS N
FOR EACH ROW
BEGIN
UPDATE WORKIN
SET ____(F)____
WHERE ____(G)____
END

7. Please answer the following questions:

Suppose there is a Electronic Toll Collection (ETC) related to the highway:

When a vehicle drives out of the station, firstly, we read the current balance X from its credit card A, and we records it as X = R(A); Then we deduct the cost S and we records it as X = X - S; Finally, we write the balance X into credit card A, and we records it as W(A,X). We also assume that more than one vehicles can use the same credit card at the same time.

Please answer the following questions based on the above assumptions:

(1) If two vehicles T1 and T2 bound to the same credit card A drive out of the cost station at the same time, their order execution sequence is as follows:

T1: X1 = R(A),X1 = X1 - S1, W(A,X1),

T2: X2 = R(A),X2 = X2 - S2, W(A,X2),

What problems will occur at this time? Could you explain why? (5)

(2) To solve the above problems, the exclusive Lock XLock(A) is introduced to lock data A, and the UnLock UnLock(A) is introduced to unlock data A. Please supplement the above execution sequence to make it meet the 2PL protocol. (5)

(3) When two vehicles bound to the same credit card number pass through the toll booth at the same time, the possible sequence of instruction executions is: x1=R(A), x1 =x1-a1, x2 = R(A), x2 = x2-a2, W(A， x1), W(A， x2). To solve the above problem, the exclusive lock instruction XLock(A) is introduced to lock the data A, and the unlock instruction Unlock(A) is used to unlock the data A. Please supplement the above execution sequence to satisfy the 2PL protocol.(5)

Answer

1． 什么是数据库的完整性？它和数据库的安全性有什么不同？

1. What is database integrity? How is it different from database security?

Reference Answer：

The integrity of the database refers to the correctness, validity and compatibility of the data, and its purpose is to protect the data from the damages caused by the legitimate users unintentionally.

Different from the integrity， the security of database is to protect data from damages caused by illegal users intentionally.

参考资料1：摘自《施伯乐老师教材》

数据库的完整性是指数据的正确性、 有效性和相容性，目的防止错误数据进入数据库。其中正确性是指数据的输入类型是否合法；有效性是指数据的值是否属于定义的有效范围；相容性是指表示同一事实的两个数据是否是一致的， 不一致就是不相容。

安全性是保护数据以防止非法用户故意造成的破坏；

完整性是保护数据以防止合法用户无意中造成的破坏；

参考资料2：摘自《王能斌老师教材》

数据库的安全性是针对"人为的破坏，比如数据被不该知道的人访问、篡改甚至破坏"的情况；

数据库的完整性是针对"输入或者更新数据库的数据有误，更新事务为遵从保持数据库一致性原则"的情况。

参考资料3：摘自《网络》

① 数据库的完整性是为了在数据的添加、删除、修改等操作中不出现数据的破坏或多个表数据不一致，是指存储在数据库中的数据正确无误并且相关数据具有一致性，数据在逻辑上的一致性、正确性、有效性和相容性。

② 数据库的安全性是指保护数据库以防止不合法使用所造成的数据泄露、更改或破坏 。

③ 数据的完整性和安全性是两个不同概念，数据库的完整性是指数据的正确性和相容性。数据库的安全性是指保护数据库，以防止他人不合法的使用造成的数据泄密、篡改或破坏。

2.数据库恢复的基本原则是什么？具体实现方法是什么？

2.What is the basic principle of database recovery? What is the implementation of database recovery?

Reference Answer：

The basic principle of database recovery is Redundancy.

The specific implementation of database recovery is Dumping Data and Recording Log Files. Data dumping refers to periodically copying the data of the entire database to

another storage medium; Log files recording is to record the start and end of each transaction, and restore the data by redo operation or undo operation to the information based on the records in the log file.

参考资料1：摘自《施伯乐老师教材》

要使数据库具有可恢复性，基本原则很简单，就是“ 冗余”，即数据库的重复存储。

数据库恢复具体实现方法如下：

（1）平时做好两件事： 转储和建立日志。

① 周期地（比如一天一次）对整个数据库进行拷贝，转储到另一个磁盘或磁带一类存储介质中。

② 建立日志数据库。记录事务的开始、结束标志，记录事务对数据库的每一次插入、 删除和修改前后的值， 写到“ 日志”库中， 以便有案可查。

（2）一旦发生数据库故障， 分两种情况进行处理：

① 如果数据库已被破坏，例如磁头脱落、 磁盘损坏等，这时数据库已不能用了，就要装入最近一次拷贝的数据库备份到新的磁盘，然后利用日志库执行“重做”(ReDo)处理，将这两个数据库状态之间的所有更新重新做一遍。 这样既恢复了原有的数据库， 又没有丢失对数据库的更新操作。

② 如果数据库未被破坏，但某些数据不可靠，受到怀疑。例如程序在批处理修改数据库时异常中断。 这时不必去拷贝存档的数据库。 只要通过日志库执行“撤销”（UnDo）处理，撤销所有不可靠的修改，把数据库恢复到正确的状态。

参考资料2：摘自《王能斌老师教材》

对于恢复，数据冗余是必须的。

具体的恢复技术大致为下列三种：

（1） 单纯以后备副本为基础的恢复技术：核心是周期性地把磁盘上的数据库转储(dump)到磁带上，这样的数据库副本可以脱机存放，不受系统故障影像，成为后备副本。

（2） 以后备副本和运行记录为基础的恢复技术：通过运行记录保持恢复所必需的数据，运行记录一般不能和数据库放在同一磁盘上，因经常把运行记录复制到磁带上，防止磁盘故障导致数据“全军覆没”。

（3） 基于多副本的恢复技术：设置多个具有独立失效模式的数据库副本，这些副本可以互为备份，用于恢复。

参考资料3：摘自《网络检索材料》

1.恢复操作的基本原理：冗余

2.恢复机制的关键技术：

建立数据冗余数据的方法：数据转储，登陆日志文件

3.关系 R，S 如下图所示。请给出下列各表达式的运算结果。

R

A	B	C	D
a_1	b_1	c_1	d_1
a_1	b_1	c_2	d_2
a_1	b_1	c_1	d_3
a_2	b_2	c_1	d_1
a_2	b_2	c_2	d_2
a_3	b_3	c_1	d_1

S

C	D
c_1	d_1
c_2	d_2

(1) 计算关系代数运算： $\Pi_{A,C,D}(_{A!=a1}(R))\div S$

(2) 用域关系演算表达除法操作。

3.The relations R and S are shown in the figure below. Please provide the results of the following expressions.

R

A	B	C	D
a_1	b_1	c_1	d_1
a_1	b_1	c_2	d_2
a_1	b_1	c_1	d_3
a_2	b_2	c_1	d_1
a_2	b_2	c_2	d_2
a_3	b_3	c_1	d_1

S

C	D
c_1	d_1
c_2	d_2

(1) Calculate the relational algebra operation: $\Pi_{A,C,D}(_{A!=a1}(R))\div S$

(2) Express the division operation using domain relational calculus.

参考答案：自己整理归纳

(1)

A
a_2

(2) 设有关系 R（X，Y）和 S（Y），X 和 Y 为属性组。则域关系演算表达式如下：

$$\{X|\exists X\forall Y(R(X,Y)\ AND\ S(Y))\}$$

Reference Answer：

(1)

A
a_2

(2)Given relations R(X, Y) and S(Y), where X and Y are sets of attributes. The domain relational calculus expression is as follows:

$$\{X|\exists X\forall Y(R(X,Y)\ AND\ S(Y))\}$$

4.请回答以下题目：

随着世界杯的召开，如何管理好足球活动成为热门话题，针对这类问题我们构建了一个足球活动管理系统，该系统需要维护球队、球员、主教练、主裁判、赞助商、单场比赛等信息，其数据模式设置如下：

·球队的信息表(球队编号，球队名称，球队人数，成立时间，主教练)

·球员的信息表(球员编号，球员姓名，出生日期，身高，体重)

·主教练的信息表(教练编号，教练姓名，出生日期，教练级别)

·主裁判的信息表(裁判编号，裁判姓名，出生日期，裁判级别)

·赞助商的信息表(赞助商编号，赞助商名称)

·单场比赛的信息表(比赛场次编号，主队编号，客队编号，主裁判身份证号，比赛日期，比分)

假设球队活动同时满足以下安排：

[1] 每个球队有一个主教练和多名球员；

[2] 每个主教练只能受聘于一个球队，每个球员只能效力于一个球队；

[3] 每场比赛都有两个球队参赛，且其中一个为主队，一个为客队，每场比赛只要一个主裁判，每场比赛都有唯一的赛事编号并且都记录了比赛日期和比分情况。

[4] 每个赞助商可以支持多个球队，但每个球队只能有一个赞助商。每个赞助商可以单独赞助多个球员，每个球员也可以为多个赞助商代言。

[5] 补充声明，存在球队单日参加多场比赛或者单日与同一队伍进行多次比赛的情况。

4.Please answer the following questions:

With the coming of the World Cup, how to manage football activities has become a hot topic. In order to solve this problem, we built a <u>Football Activity Management System</u>, which needs to maintain the information of the <u>Team</u>, <u>Player</u>, <u>Chief Coach</u>, <u>Chief Referee</u>, <u>Sponsor</u>, And each <u>Match</u>. The data schema is set as follows:

[1] Team (TeamId, TeamName, PlayersNum, EstablishDate, ChiefCoach) ;

[2] Player (PlayerId,PlayerName, BirthDate, Height, Weight) ;

[3] Chief Coach (CoachId,CoachName, BirthDate, CoachLevel) ;

[4] Chief Referee (RefereeId,RefereeName, BirthDate, RefereeLevel) ;

[5] Sponsor (SponsorId, SponsorName)

[6] Match (MatchId, HomeTeamId, VisitTeamId, ChiefRefereeID, MatchDate, Score).

Assume that the football system meet the following arrangements:

[1] Each team has only one chief coach and several players.

[2] Each chief coach can only be hired by one team, and each player only play for one team.

[3] There are two teams in each match, one for the home team and one for the visiting team. There is only one chief referee for each match. Each match has a unique id and the date and score of the match are recorded.

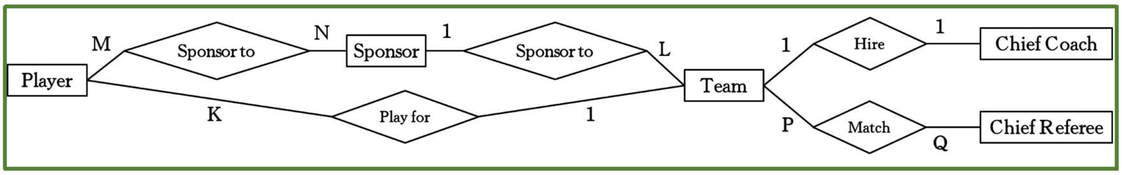
[4] Each sponsor can sponsor multiple teams, but each team can only have one sponsor. Each sponsor can individually sponsor multiple players, and each player can also represent multiple sponsors.

[5] Additionally, it allows the cases where one team plays more than one matches in the same day or plays with the same team in the same day.

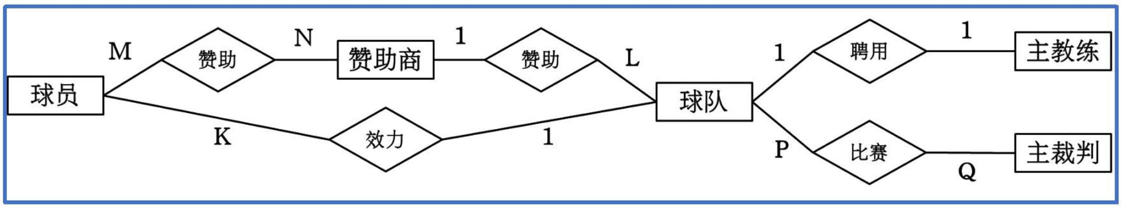
（1）请根据上述的设置绘制出改足球活动管理系统的ER图.

（1）Please draw the ER diagram of the football activity management system according to the above Settings).

Reference Answer:



参考答案1：自己整理归纳



（2）请写出单场比赛的信息表中的主键、候选键和外键

(2) Please write the Primary Key、 Candidate Key and Foreign Key of the Match Table.

Reference Answer:

Primary key: {MatchId}
Candidate key: {MatchId}
Foreign key: {HomeTeamId, VisitiTeamId,ChiefRefereeId}

参考答案1：自己整理归纳

主键：{比赛场次编号}
候选键：{比赛场次编号}
外键：{主队编号，客队编号，主裁判身份证号}

5.请回答以下题目：

有教师和课程的关系如下表所示：

课程号(候选键)	教师姓名	部门编号
C1	David	D1
C2	Mally	D2
C3	Musk	D1
C4	Musk	D1

5. Please answer the following questions:

The relationship between Teachers and Courses is shown in the following table:

Course ID(key)	Teacher Name	Department ID
C1	David	D1
C2	Mally	D2
C3	Musk	D1
C4	Musk	D1

（1）请问该课程-教师关系属于第几范式？为什么？

(1) Which paradigm does the above relationship belong to? Why?

Reference Answer：

It's 2NF.

In this table, the primary attribute is the Course ID, and the non-primary attributes are the Teacher Name and Department ID.

And there is the following relationship: Course ID → Teacher Name, Teacher → Department ID, so we get Course ID → Department ID;

Firstly, each attribute value in the relation is an atomic value that cannot be subdivided, so it satisfies 1NF.

Secondly, every non-primary attribute value in the relation completely depends on the candidate key Course, so it satisfies 2NF.

Finally, there is a transitive dependency of the non-primary attribute Teacher Department on the candidate key Course, so it does not satisfy 3NF.

参考答案1：自己整理归纳

属于2NF。

在该表中，主属性是Course ID，非主属性是Teacher Name 和 Department ID;

并且存在以下关系：Course ID → Teacher Name ， Teacher → Department ID，因此有 Course ID → Department ID;

首先该关系中的每个属性值都是不可再分的原子值，因此满足1NF;

其次该关系中的每个非主属性值都完全函数依赖候选键Course,因此满足2NF;

最后，因为存在非主属性Teacher Department对候选键Course的传递依赖情况，因此不满足3NF;

（2）请问是否存在"删除异常"和"修改异常"问题？如果存在，请说明分别是怎样发生的？

(2) Does there exist the problem of "Delete Exception" or "Modify Exception"? If it does, how do they happen?

Reference Answer：

It exists.

"Deletion exception" : Occurs when the deletion of a course information lead to the loss of the teacher information.

"Modification exception" : occurs when a teacher leaves the original department to another department, and all records of the teacher need to be modified.

参考答案1：自己整理归纳

存在。

"删除异常"：发生在当删除某门课程信息的时候会导致教师信息也被删除而丢失的情况。

"修改异常"：发生在当一个教师离开原本部门去到其他部门时，需要修改该老师的所有记录。

6.请回答以下题目：

已知某地区的地方企业数据关系模式如下：

员工的信息表EMP(员工工号，员工姓名，员工性别，员工年龄，员工职称)

企业的信息表CPY(企业编号，企业名称，所在辖区)

请根据以上信息回答问题：

6. Please answer the following questions:

It is known that the data relationship of local enterprises is as follows:

[1] Information of Employees: EMP(Eno, Ename, Sex, Age, Title)

[2] Information of Companys：CPY(Cno, Cname, Area)

Please answer the question according to the above information:

（1） 请创建一个员工在企业就职的信息表WORKIN，并满足以下条件：

[1] 信息表中包含员工工号、企业编号、当前工资；

[2] 指定主键和外键信息；

[3] 设置工资不能低于3000元的约束限制。

(1) Please CREATE an information table WORKIN where employees work in the enterprise and meet the following conditions:

[1] The WORKIN table contains three attribute (the Eno from Emp, the Cno from CPY and the Salary)

[2] Define the primary and foreign key.

[3] Set a constraint that the Salary should not be less than 3,000 yuan.

Reference Answer:

```
CREATE TABLE WORKIN(
    Eno CHAR(10) REFERENCES EMP(Eno),
    Cno CHAR(10) REFERENCES CPY(Cno),
    Salary int CHECK(Salary>=3000),
    PRIMARY KEY Eno,Cno
)
```

参考答案1：自己整理归纳

```
CREATE TABLE WORKIN(
    Eno CHAR(10) REFERENCES EMP(Eno),
    Cno CHAR(10) REFERENCES CPY(Cno),
    Salary int CHECK(Salary>=3000),
    PRIMARY KEY Eno,Cno
)
```

（2） 基于问题(1)中创建的WORKIN表格，填写以下SQL语句的实现功能：

a) 查询员工最多的公司编号和公司名称

SELECT C.Cno,C.Cname
FROM Company AS C，WORKIN AS W
WHERE C.Cno = W.Cno
GROUP BY ____(A)____
HAVING ____(B)____ （SELECT COUNT(*)
FROM WORKIN AS W2
GROUP BY W2.Cno）

(2) Based on the WORKIN table created in question (1), fill in the following SQL statement implementation function:

a) Query the Cno and Cname of the company which has the most employees

SELECT C.Cno,C.Cname
FROM Company AS C，WORKIN AS W
WHERE C.Cno = W.Cno
GROUP BY ____(A)____
HAVING ____(B)____ （SELECT COUNT(*)
FROM WORKIN AS W2
GROUP BY W2.Cno）

Reference Answer:

（A） C.Cno, C.Cname

（B） COUNT(*) >= ALL

参考答案1：自己整理归纳

（A） C.Cno, C.Cname

（B） COUNT(*) >= ALL

b) 查询员工平均年龄最大的企业编号和所在辖区

SELECT C.Cno，C.Area
FROM EMP AS E，CPY AS C，WORKIN AS W
WHERE W.Cno = C.Cno
AND W.Eno = E.Eno
GROUP BY C.Cno, C.Area
HAVING ____(C)____ >= ____(D)____

b) Query the Cno and Area of the Company whose employees have the oldest average Age

SELECT C.Cno，C.Area
FROM EMP AS E，CPY AS C，WORKIN AS W
WHERE W.Cno = C.Cno
AND W.Eno = E.Eno
GROUP BY C.Cno, C.Area
HAVING ____(C)____ >= ____(D)____

Reference Answer：

（C） AVG(E.Age))
（D） (SELECT MAX(Avgage)
FROM (SELECT AVG(E2.Age) AS Avgage
FROM EMP AS E2，CPY AS C2，WORKIN AS W2
WHERE W2.Cno = C2.Cno
AND W2.Eno = E2.Eno
GROUP BY C2.Cno, C2.Area
)
）

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（C） AVG(E.Age))
（D） (SELECT MAX(Avgage)
FROM (SELECT AVG(E2.Age) AS Avgage
FROM EMP AS E2，CPY AS C2，WORKIN AS W2
WHERE W2.Cno = C2.Cno
AND W2.Eno = E2.Eno
GROUP BY C2.Cno, C2.Area
)
）

(3)请填写以下触发器的空缺部分来实现员工工资的自动维护功能，要求当员的职称发生变化的时候，需要根据“职称-收入函数”SalaryList(Title)计算其新的职称对应工资值并进行自动更新：

```
CREATE TRIGGER NewSalary
AFTER UPDATE ____(E)____ON EMP
REFERENCING NEW AS N
FOR EACH ROW
BEGIN
    UPDATE WORKIN
    SET ____(F)____
    WHERE ____(G)____
END
```

(3) Please fill in the vacancy part of the following TRIGGER to realize the automatic maintenance function of employee salary. When the Title of the employee changes, the salary of the new title should be calculated according to the "Title-Salary" Function——SalaryList(Title) . And the salary should be updated automatically:

```
CREATE TRIGGER NewSalary
AFTER UPDATE ____(E)____ON EMP
REFERENCING NEW AS N
FOR EACH ROW
BEGIN
    UPDATE WORKIN
    SET ____(F)____
    WHERE ____(G)____
END
```

Reference Answer：

（E）OF Title

（F）Salary = SalaryList（N.Title）

（G）Eno = N.Eno

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（E）OF Title

（F）Salary = SalaryList（N.Title）

（G）Eno = N.Eno

7.请回答以下题目：

假设有个与高速路不停车收费系统(ETC)：

当车辆开驶出收费站点时，读取其信用卡A的当前余额X，记X=R(A);然后扣除过路费S，记X = X - S ；最后将扣费后的余额X写入信用卡A，记W(A,X)。此外假设多个车辆可以绑定同一个信用卡。

请根据上述假设回答以下问题：

7. Please answer the following questions:

Suppose there is a toll collection system (ETC) related to the highway:

When a vehicle drives out of the station, firstly, we read the current balance X from its credit card A, and we records it as X = R(A); Then we deduct the cost S and we records it as X = X - S; Finally, we write the balance X into credit card A, and we records it as W(A,X). We also assume that more than one vehicles can use the same credit card at the same time.

Please answer the following questions based on the above assumptions:

（1） 如果两个绑定同一信用卡A的车辆T1和T2同时开出收费站,其指令执行序列如下:

T1 : X1 = R(A) , X1 = X1 – S1， W(A,X1),

T2 : X2 = R(A) , X2 = X2 – S2 , W(A,X2),

请问此时会出现什么问题？请解释其原因？

(1) If two vehicles T1 and T2 bound to the same credit card A drive out of the cost station at the same time, their order execution sequence is as follows:

T1: X1 = R(A),X1 = X1 - S1, W(A,X1),

T2: X2 = R(A),X2 = X2 - S2, W(A,X2),

What problems will occur at this time? Could you explain why?

Reference Answer:

There will be lost updates.
Because the balance value read by vehicle T2 is not yet written back by T1, the balance value written by vehicle T1 in the payment will be overwritten by vehicle T2, so that the balance cannot be withdrawn from vehicle T1's payment, which belongs to the database inconsistency caused by the loss and modification.

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会出现更新丢失的情况,。
因为车辆T2读取的余额值是T1尚未写回的，导致车辆T1在缴费写入的余额值会被车辆T2的覆盖，使得余额不能提现车辆T1的缴费情况，属于丢失修改造成的数据库不一致性。

（2）为解决以上问题，引入独占指令锁XLock(A) 对数据A进行加锁，解除指令锁UnLock(A)对数据A进行解锁，请补充上述执行序列，使其满足2PL协议。
(2) To solve the above problems, the exclusive Lock XLock(A) is introduced to lock data A, and the UnLock UnLock(A) is introduced to unlock data A. Please supplement the above execution sequence to make it meet the 2PL protocol.

Reference Answer:

T1:XLOCK(A),X1=R(A),X1= X1–S1,W(A,X1),UNLOCK(A);
T2:XLOCK(A),X2=R(A),X2= X2–S2,W(A,X2),UNLOCK(A);

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T1:XLOCK(A),X1=R(A),X1= X1–S1,W(A,X1),UNLOCK(A);
T2:XLOCK(A),X2=R(A),X2= X2–S2,W(A,X2),UNLOCK(A);

考察点：
进行 W 操作前，要对修改的表加上 XLock()指令。
在进行单个游客的预订事务时，所有的 XLOCK()要在 UNLOCK()前面，以满足 2PL 规则

（3）当两个绑定到同一信用卡号的车辆同时经过收费口时，可能的指令执行序列为：x1=R(A)，x1 =x1-a1, x2 = R(A), x2 = x2-a2, W(A，x1)，W(A，x2)。为了解决上述问题，引入独占锁指令XLock(A)对数据A进行加锁，解锁指令Unlock(A)对数据A进行解锁。请补充上述执行序列，使其满足2PL协议。
(3) When two vehicles bound to the same credit card number pass through the toll booth at the same time, the possible sequence of instruction executions is: x1=R(A), x1 =x1-a1, x2 = R(A), x2 = x2-a2, W(A， x1), W(A， x2). To solve the above problem, the exclusive lock instruction XLock(A) is introduced to lock the data A, and the unlock instruction Unlock(A) is used to unlock the data A. Please supplement the above execution sequence to satisfy the 2PL protocol.

Reference Answer：

XLock(A), x1=R(A), x1=x1-a1, W(A,x1), Unlock(A), XLock(A), x2 = R(A), x2 = x2-a2, W(A,x2), Unlock(A)

参考答案1：

加锁后的执行序列：**XLock(A), x1=R(A), x1=x1-a1, W(A,x1）, Unlock(A), XLock(A), x2 = R(A), x2 = x2-a2, W(A,x2), Unlock(A)**

考察点：

进行 W 操作前，要对修改的表加上 XLock()指令。

在进行单个游客的预订事务时，所有的 XLOCK()要在 UNLOCK()前面，以满足 2PL 规则

保证读/写锁不交叉，即保证释放锁之后才能执行加锁操作。